AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 14 · STUDENT LESSON

Hallucination Failure Types

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Split the claims. Check the evidence. Repair with support.

Distinguish source contradictions, missing support, quotation mismatches and unjustified inferences.

Keep one habit: Status definitions → room audit → trip calculation → complete paragraph/source audit → independent summary.

MAIN LESSON · SECTION 1

Which claim is supported, contradicted or still unknown?

> Inspect individual claims rather than treating a whole fluent answer as correct.

Your learning goals

I can explain the **mechanism** and terminology of generation failures and verification.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall source roles and uncertainty

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Multimodal **Evidence**: distinguish a directly supported observation, an unverified caption and hidden information. A fluent generated sentence needs the same claim-level **evidence** check.

Your end goal

By the end, I can distinguish supported, **contradicted** and **unsupported** generated claims.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Multimodal **Evidence**. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Structured **Output** and Schemas. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 14: Hallucination Failure Types

Fluent **output** may contain **unsupported claim**s, fabricated detail, or **faulty inference**.



CREATOR MISSION

Tag claims in an answer and classify each failure type.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Separate a claim from the wording that makes it sound convincing

> Verification needs the exact source or calculation, not confidence or style.

The generation-failure and verification idea

Generated answers may invent facts, misread quantities or cite irrelevant passages. Verification checks each claim against **evidence** rather than accepting fluent wording.

Use the supplied evidence within its scope

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use hallucination failure types to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

MAIN LESSON · SECTION 3

Split the answer into claims and label each status

> Keep sources, assumptions and intermediate calculations beside the result.

The atomic-claim procedure

Split an answer into **atomic claim**s. For each, record the exact source passage, support status and any needed calculation. Mark **unsupported** and **contradicted** claims separately; do not repair them by inventing sources.

Read the three distinct evidence statuses

Status	Question to ask	What it does not imply
Supported	Does the supplied passage or valid calculation establish this exact claim?	Not a guarantee that the source is correct in the world
Contradicted	Does it conflict with the supplied evidence under the stated assumptions?	Not proof of a model's internal cause or intention
Unsupported	Is the required evidence missing or the inference unjustified?	Not proof that the opposite claim is true

State units and the input contract

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the deciding evidence

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

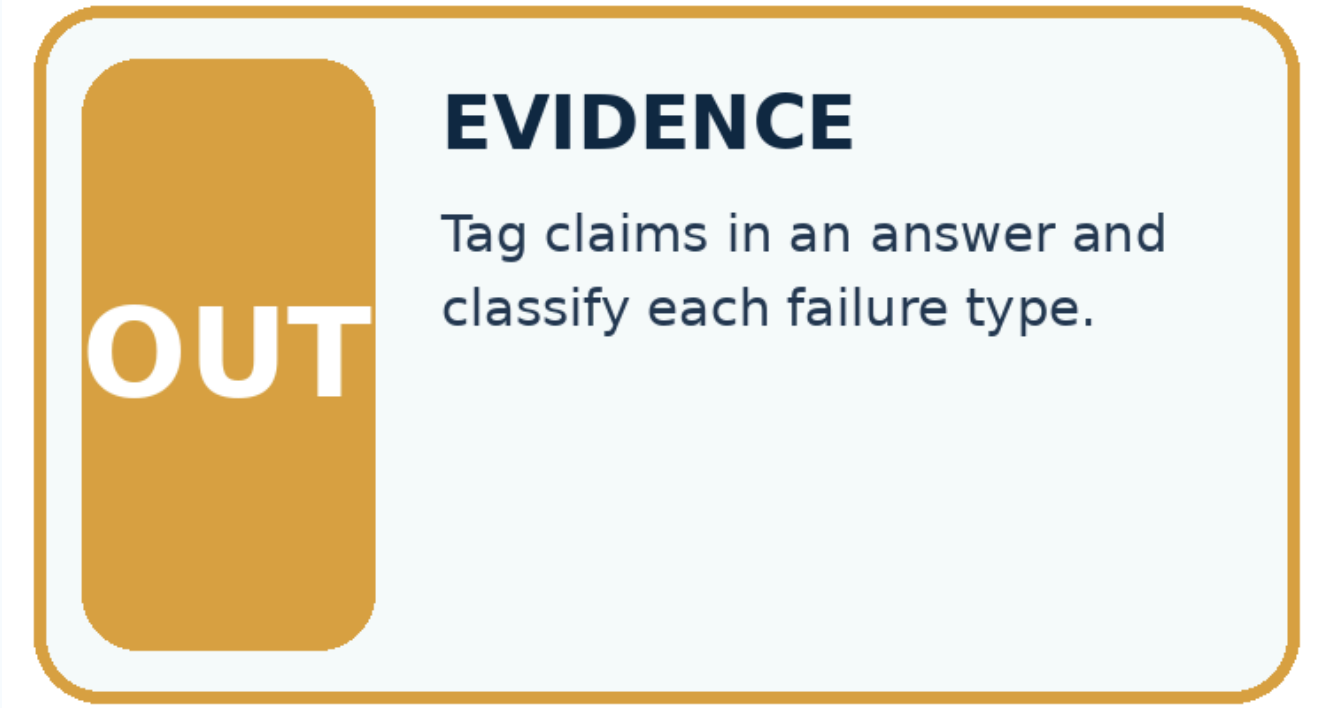
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Tag claims in an answer and classify each failure type.
↓
SYSTEM
Fluent output may contain unsupported claims, fabricated detail, or faulty inference.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Generation failures and verification — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Source R says room B seats 12 and the meeting starts at 15:00. Generated answer says B seats 20, starts at 15:00 and includes free lunch.	Split an answer into atomic claims. For each, record the exact source passage, support status and any needed calculation. Mark unsupported and contradicted claims separately; do not repair them by inventing sources.	Capacity 20 is contradicted; 15:00 is supported; free lunch is unsupported. One of three claims has support. The missing lunch fact is not evidence of no lunch.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

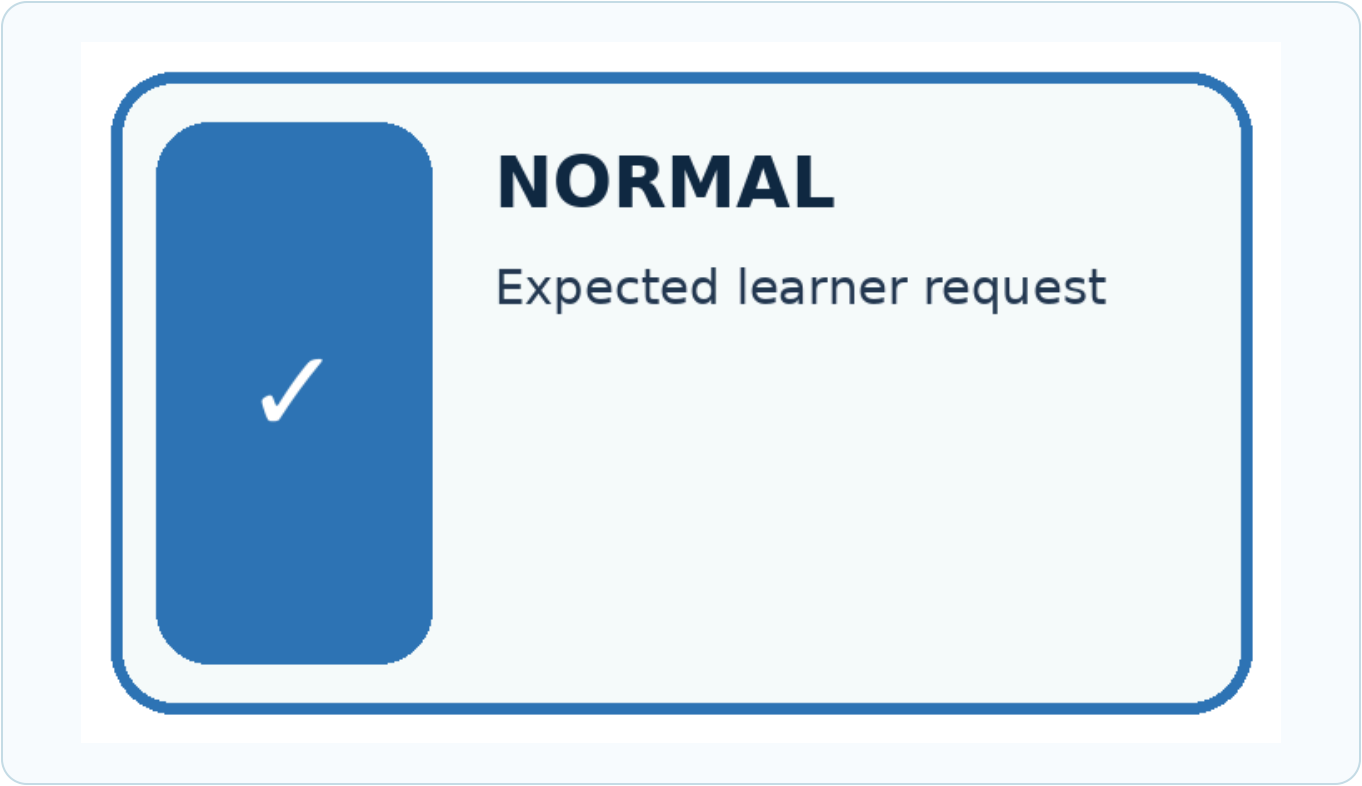
Connect the picture and the explanation

MAIN LESSON · SECTION 4

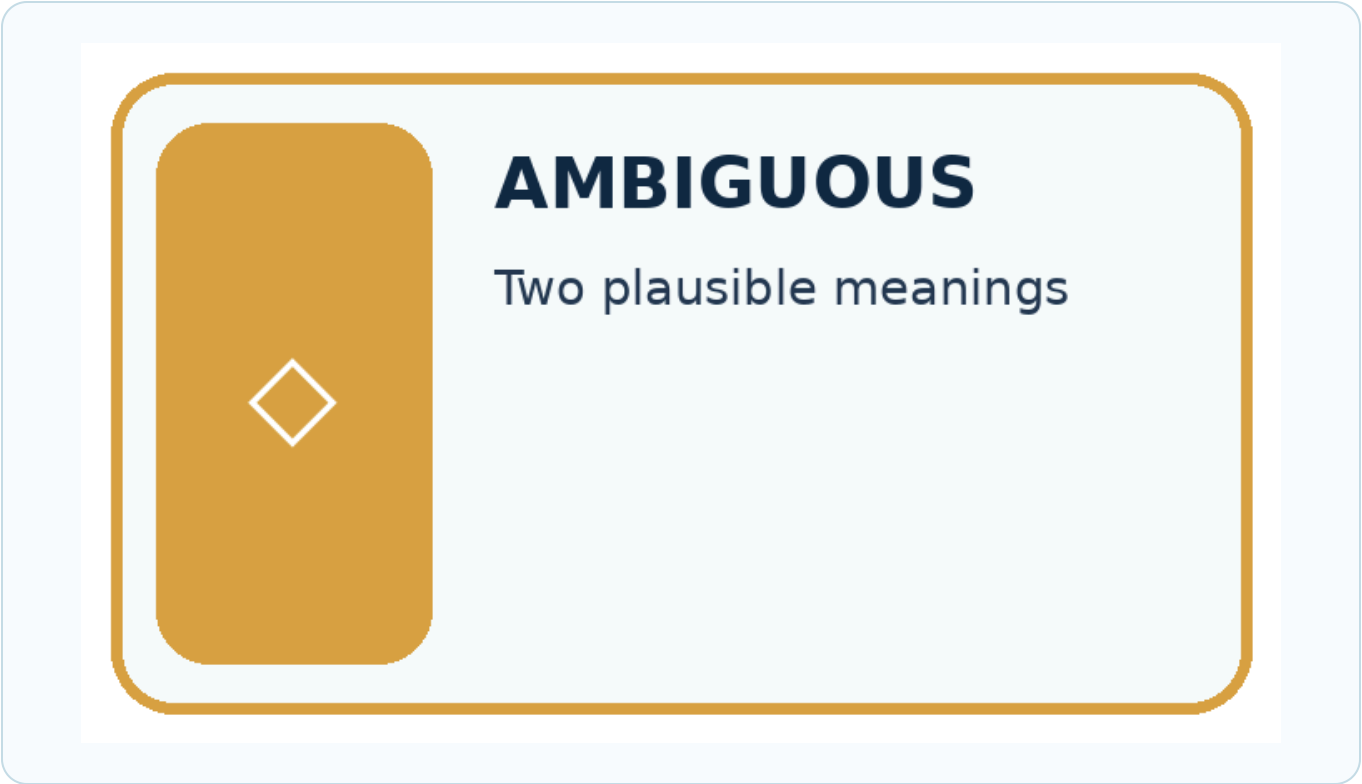
Compare missing evidence with contradictory evidence

> An unverified lunch claim is different from a contradicted duration.

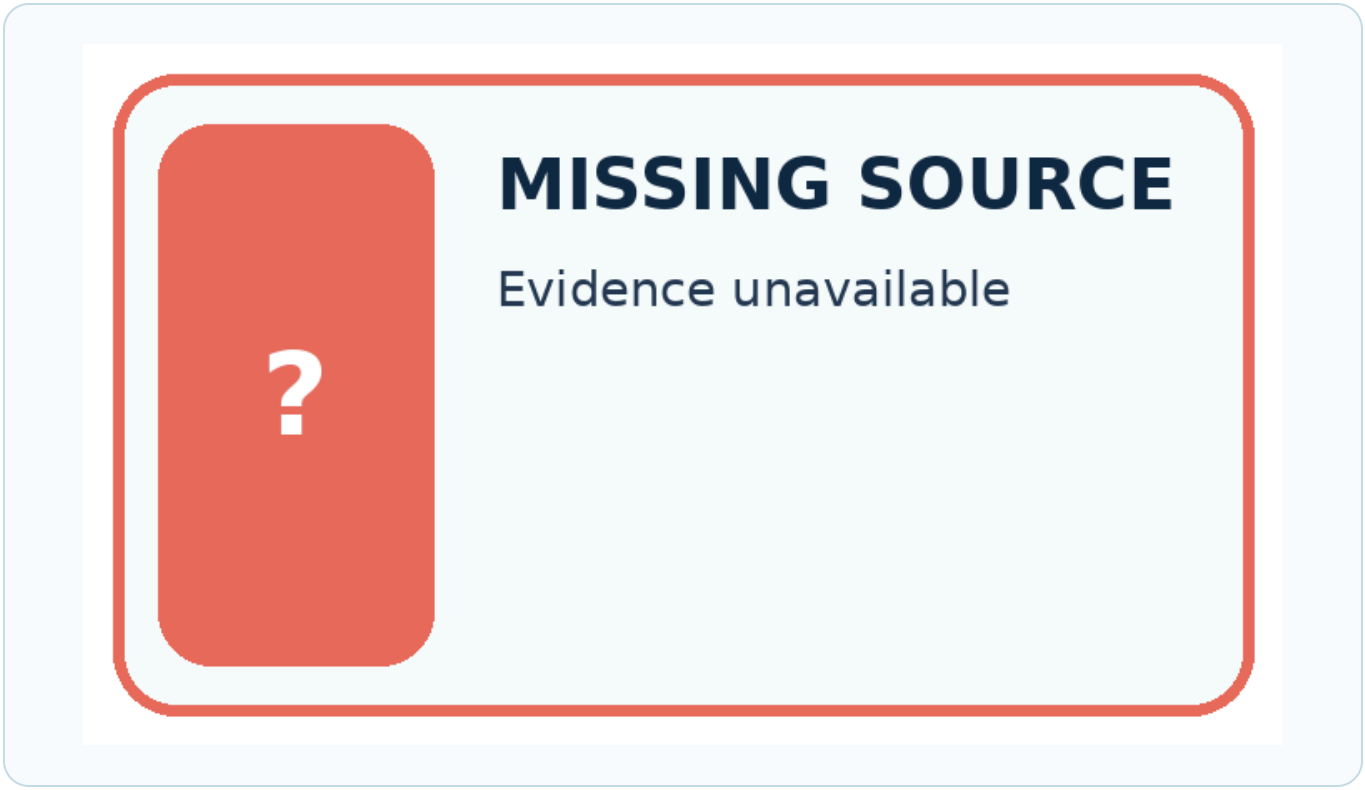
Four pictures with a purpose



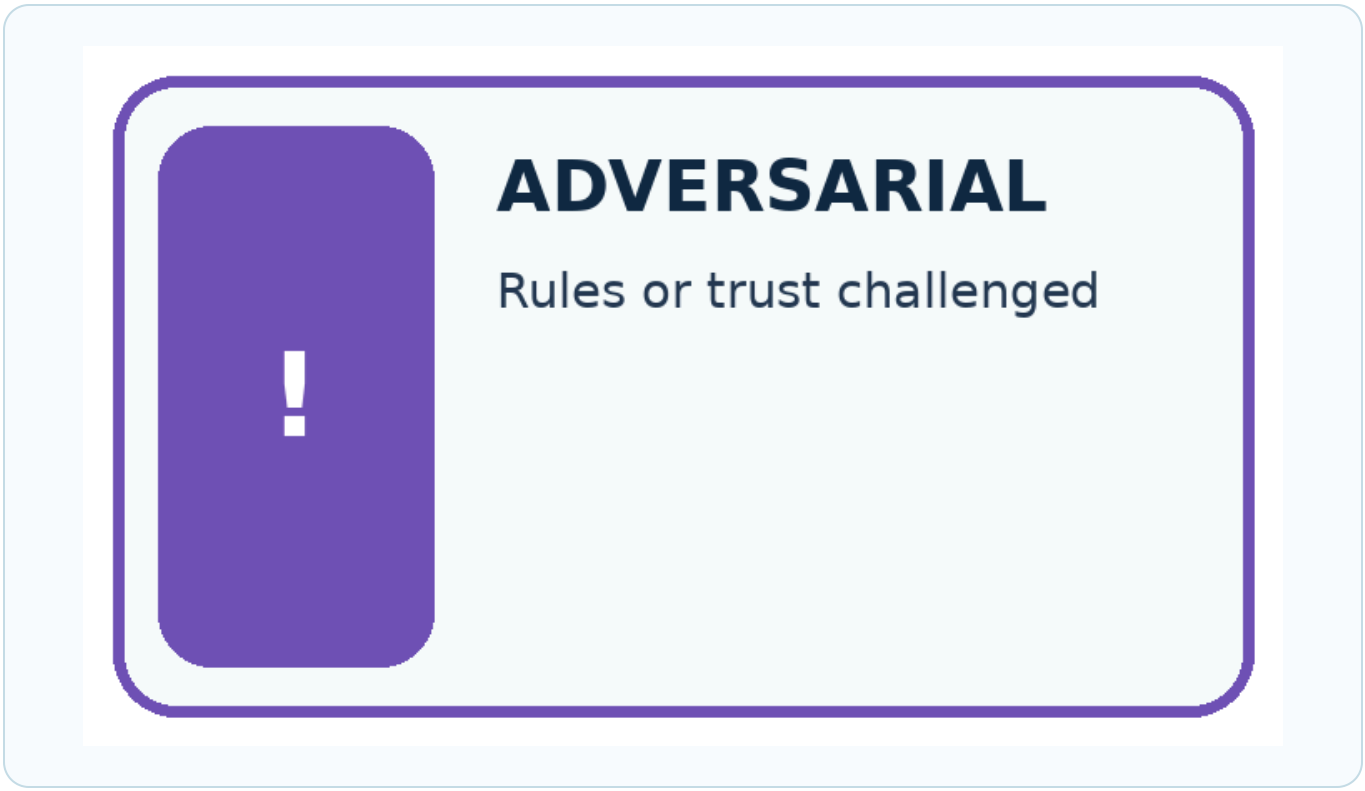
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Add a planned 30-minute stop while preserving speed and distance.

▼ Check after predicting

Prediction: Planned total elapsed time becomes 3.5 hours.

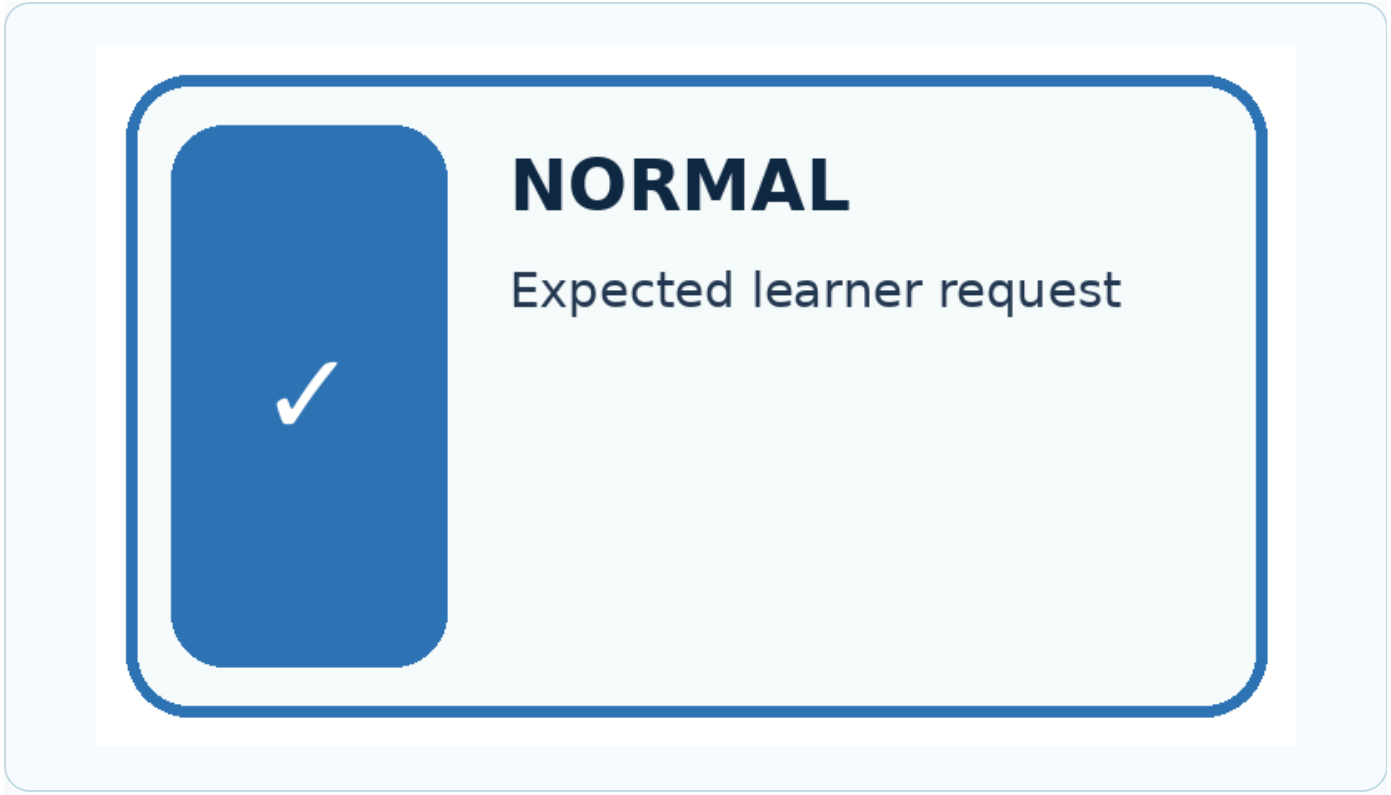
Reason: Three moving hours plus half an hour stopped; walking-time calculation itself stays three.

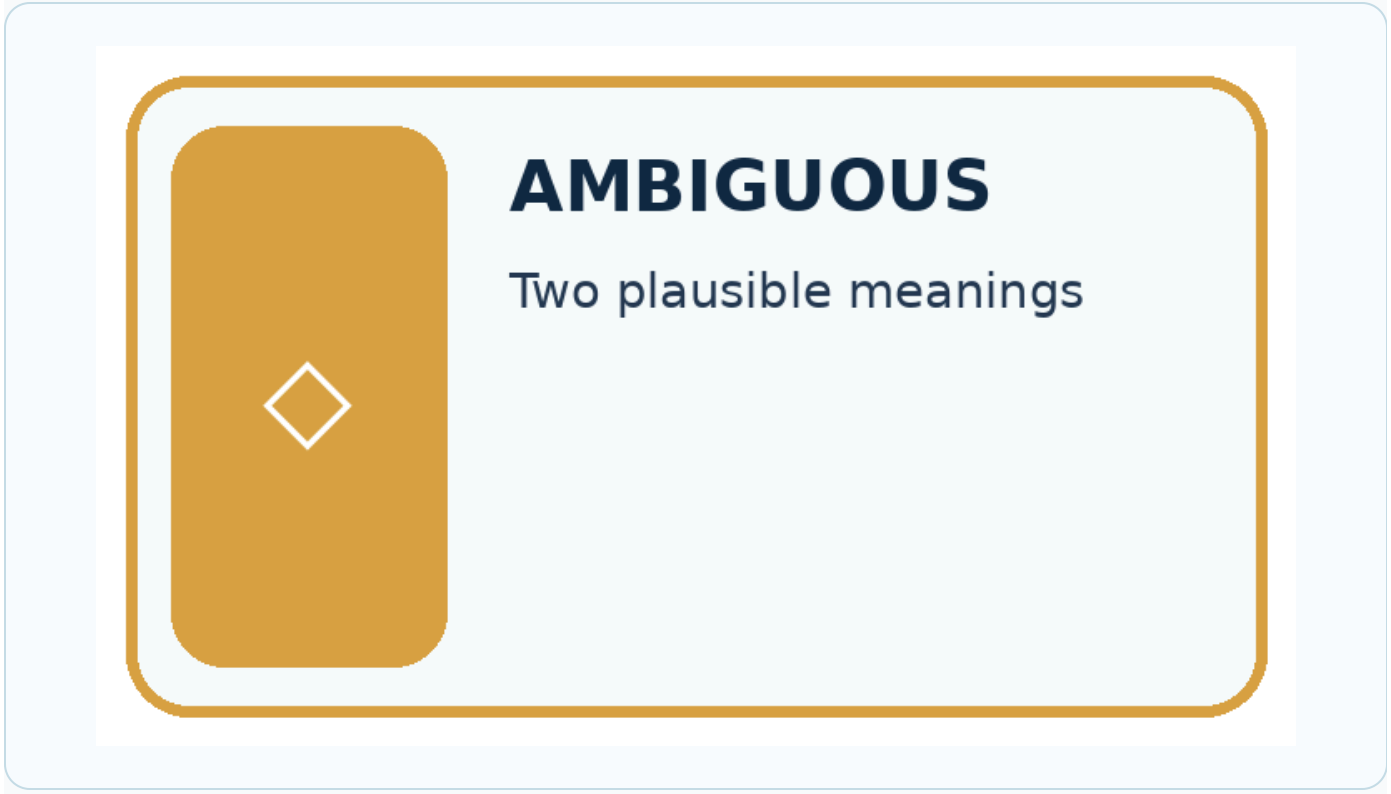
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Audit the room report, then a separate trip calculation

> Keep source R, source S and their claim counts distinct.

Read the complete room source and answer

Source R says room B seats 12 and the meeting starts at 15:00. Generated answer says B seats 20, starts at 15:00 and includes free lunch.

Check its three claim statuses

Capacity 20 is **contradicted**; 15:00 is supported; free lunch is **unsupported**. One of three claims has support. The missing lunch fact is not **evidence** of no lunch.

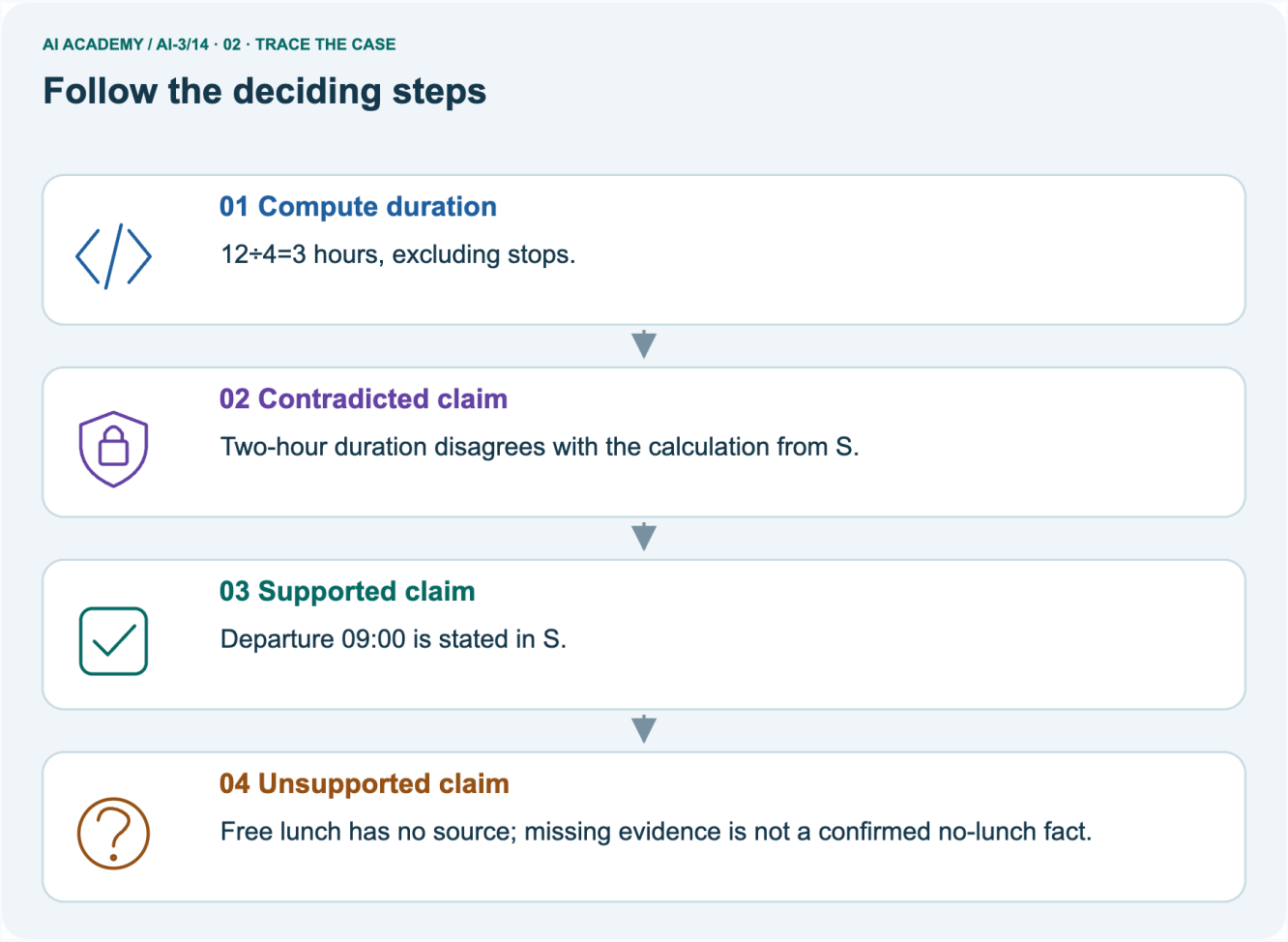
Read the separate trip plan and generated summary

A fictional Study Buddy drafts a field-trip summary from one approved record.

Source S: trail length 12 km; planned constant speed 4 km/h, excluding stops; departure 09:00.

Draft: “The walk takes two hours, starts at 09:00 and includes free lunch.” No lunch record is supplied.

Trace units, calculation and claim support



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute duration	12÷4=3 hours, excluding stops.
2. Contradicted claim	Two-hour duration disagrees with the calculation from S.
3. Supported claim	Departure 09:00 is stated in S.
4. Unsupported claim	Free lunch has no source; missing evidence is not a confirmed no-lunch fact.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

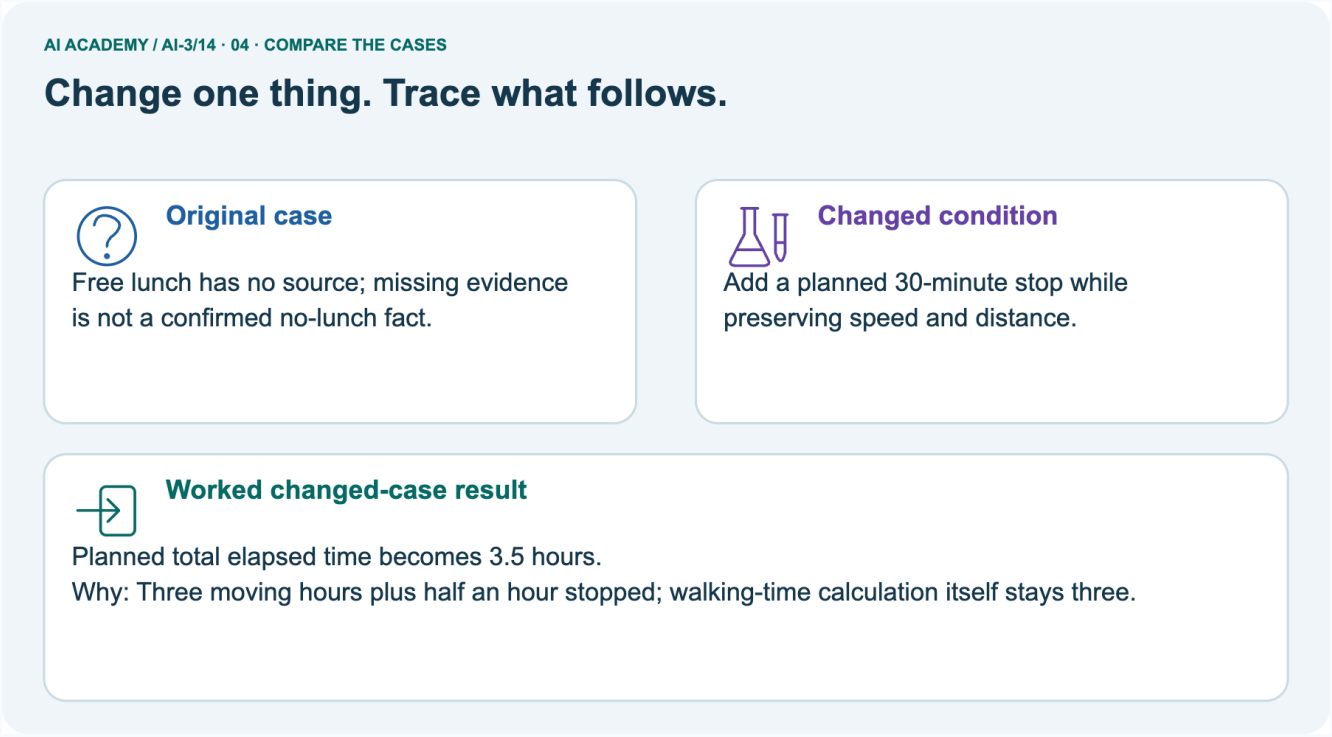
Different errors need different repairs. Recalculate the duration; keep the supported departure; remove or seek **evidence** for lunch. These categories are more informative than calling the whole answer wrong.

Arithmetic inherits assumptions such as constant speed and no stops, so avoid turning the three-hour walking calculation into a guaranteed arrival time.

Predict adding a stop lasting30minutes

- Change one thing: Add a planned 30-minute stop while preserving speed and distance. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State planned-versus-observed limits

- The fictional plan does not establish actual speed, delays or what eventually occurred on a real trip.
- Read the labelled picture: Can we conclude there will definitely be no lunch from S alone?

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 14 • WORKED EVIDENCE

1. Claim 1 is directly supported by the passage.
2. Claim 2 invents a scientist’s quotation: fabrication.
3. Claim 3 draws a cause from correlation: faulty inference.
4. Tag each claim, repair only with evidence, or mark unknown.

1. Claim 1 is directly supported by the passage.
2. Claim 2 invents a scientist’s quotation: **fabrication**.
3. Claim 3 draws a cause from correlation: **faulty inference**.
4. Tag each claim, repair only with **evidence**, or mark unknown.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

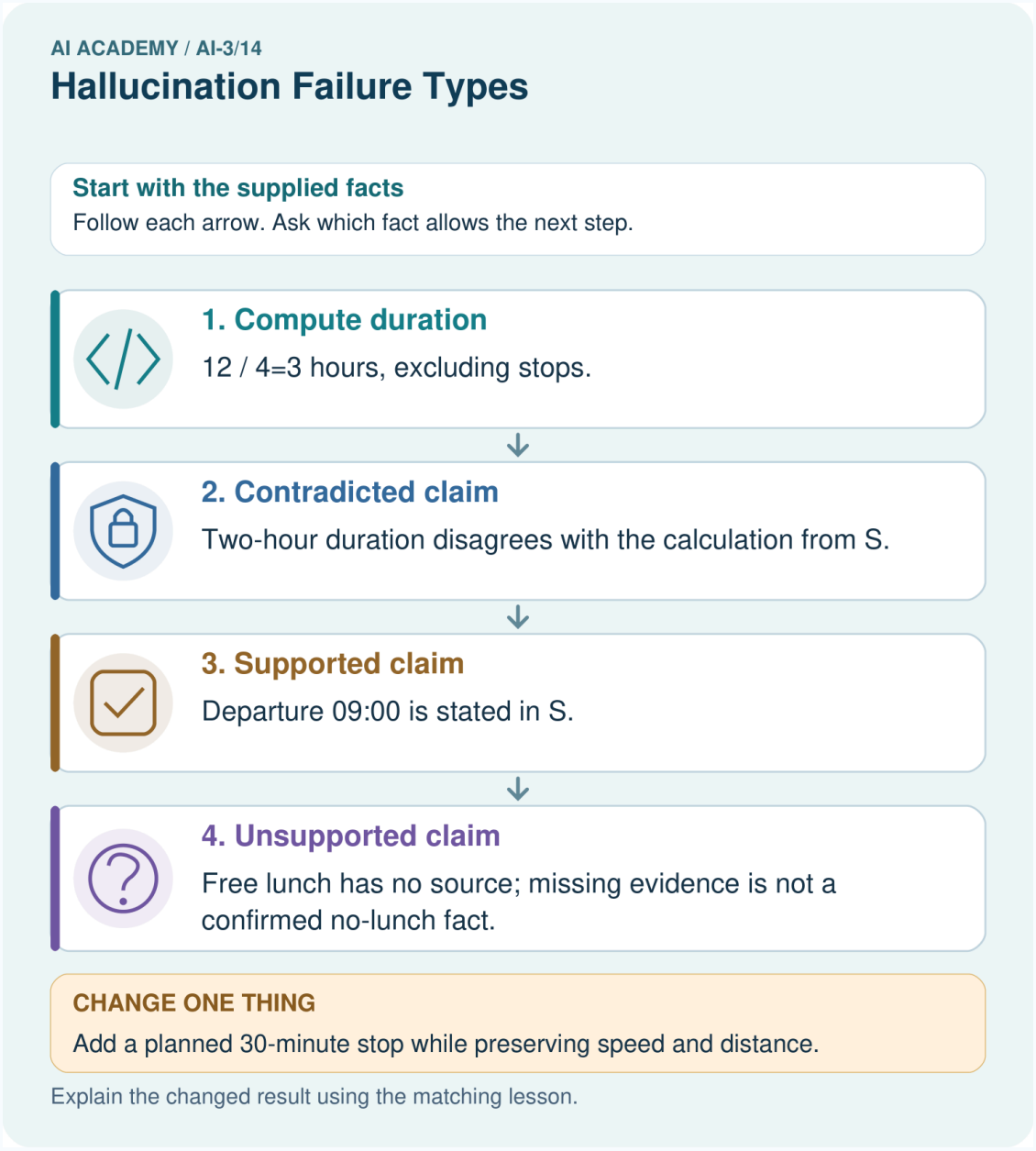
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Hallucination Failure Types

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Repair the first unsupported step without inventing a new source

> Different evidence failures require different repairs.

Find the missing-to-opposite inference

Claim to repair: The source does not mention lunch, so there definitely will be no lunch. The available evidence establishes neither free lunch nor no lunch. Mark it unverified and seek the relevant record if the question needs it. By contrast,12km at4km/h gives3moving hours under the supplied assumptions, so a2-hour calculation conflicts with those inputs. Keep the unsupported and contradicted categories separate.

Propose a revealing new check

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest **unsupported** assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A confident, detailed answer is less likely to hallucinate. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise the room report with less help

> A correct source value and a valid calculation are separate checks.

State the missing-evidence limit

Source agreement and arithmetic validity are separate checks; missing **evidence** is not proof of the opposite claim.

Complete main room-report W2

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Generation failures and verification”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair an all-future-cases claim

- Use the worked case for Generation failures and verification. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the separate trip’s lunch limit

- Can we conclude there will definitely be no lunch from S alone?

Explain the picture

My explanation using a pictured step and the given facts

Keep the independent trip case withheld

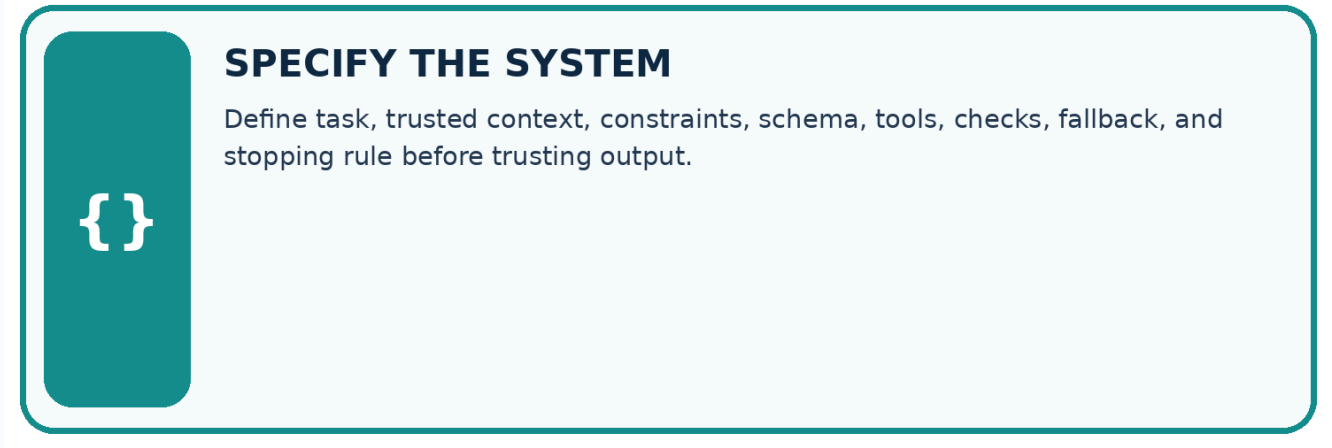
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Audit a complete paragraph against three complete sources

> The materials include factual, quotation and inference failures.

The complete approved records

Approved fictional source	Complete relevant record
R	Room B seats12. The meeting starts at15:00. No lunch information is supplied.
Q	Complete quoted line for this exercise: Check the current record.
C	Invented session L used illustrated notes and scored4/5. Session M used plain notes and scored2/5. Participants and questions differed; this was not a controlled comparison.

The complete paragraph to split

Complete fictional generated paragraph to audit: Room B seats20. The meeting starts at15:00. Free lunch is included. Q says “I guarantee free lunch.” Session L scored4/5. Illustrated notes caused better learning.

Classify, justify and repair each claim

1. Split the paragraph into six atomic claims before checking. For each identify the precise R, Q or C passage, classify supported/contradicted/unsupported, describe the failure type where supported by evidence, and write a bounded repair. For Q check the exact supplied quotation rather than inventing an absent quote source.

   For C distinguish the recorded score from a causal-learning claim: groups and questions changed too. Do not infer a model’s intention or hidden technical cause from an output alone. Color may help mark statuses, but also write the words so the record does not depend on color. Preserve the original paragraph and every correction. These are supplied fictional records, not measured model outputs or learning outcomes.

   Keep the independent18km/6kmh case for a first attempt.

Keep the evidence record

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

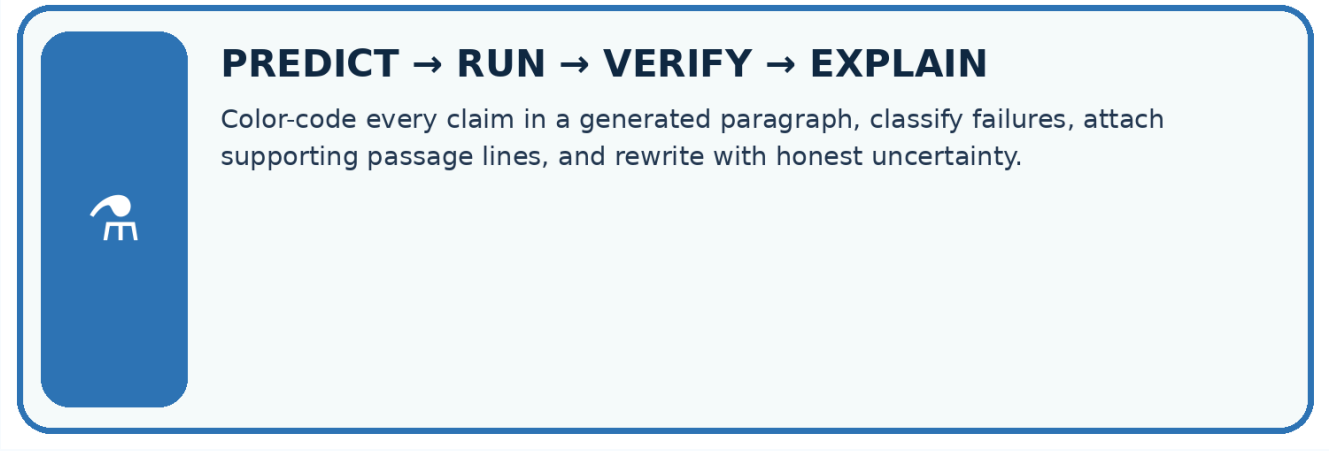
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Audit the supplied six-claim paragraph against complete R,Q,C records. Write status words as well as any color coding, inspect the exact quote and causal inference, and retain source-based repairs without inventing evidence or model intent.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit the fresh trip summary

> Use the stated assumptions and keep the three claim statuses separate.

Independent record and scope

Source S: trip is 18 km; vehicle speed is 6 km/h, constant, excluding stops. Answer says 2 hours, 18 km and includes lunch. Audit and recalculate.

Use the independent source's own distance, constant speed and exclusion of stops. Show the calculation's units and state which claims have direct, calculated or missing support. No actual trip observations or lunch record are supplied; do not import values from the worked trip or invent real conditions.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first **unsupported** step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Week 14 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

An answer gives a book title that does not exist and quotes words absent from the cited passage. Identify two failures and a repair.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Study Buddy must distinguish source-backed claims, calculations, and unverified generation.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A generated answer contains a nonexistent quote, a real source unrelated to the claim and a correct sum using the wrong **input**. Classify the three failures and their repairs.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Make each project claim traceable to its evidence

> Keep supported facts, calculations and unverified additions distinct.

This week’s concrete contribution

For the Verified Study Buddy, keep claim text, exact supporting passage/version, derived calculation and status in a visible audit record. Correct a contradicted value only from the source or a justified calculation. Remove or qualify unsupported additions rather than asserting their opposites. A supplied quotation must match its claimed source, and association alone does not establish a causal learning benefit.

Audit Study Buddy **output**s claim by claim, recording contradiction, missing support and correct calculations separately.

Keep the checkpoint and human choice

THIS WEEK'S CHECKPOINT Study Buddy must distinguish source-backed claims, calculations, and unverified generation.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

> **Write or draw your reasoning and evidence.**

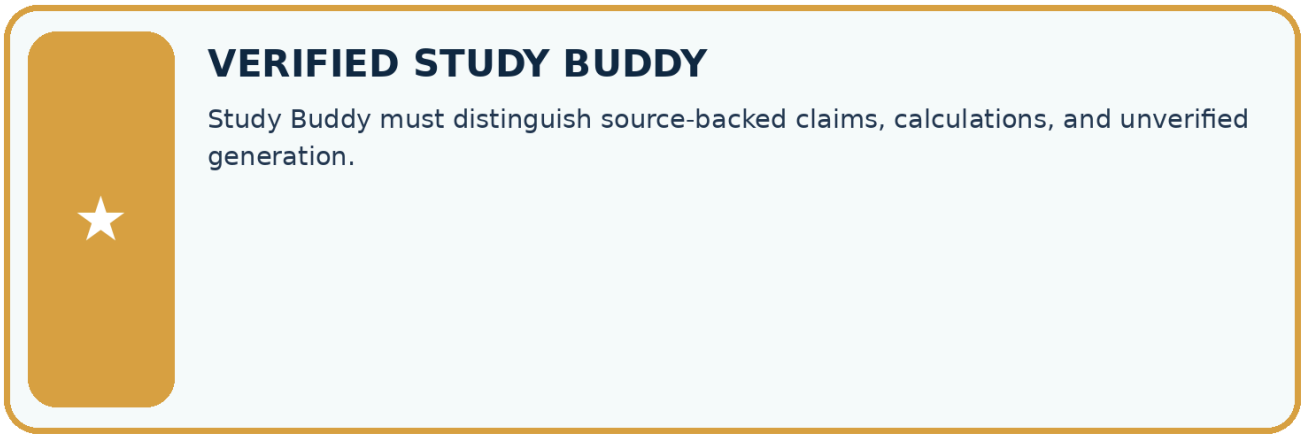
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: state the claim, deciding evidence and honest repair

> Unknown is different from false, and a plan is different from an observation.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of generation failures and verification

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
unsupported claim	A statement for which the supplied evidence does not establish the asserted conclusion.	
fabrication	Invented information presented as if it were established fact.	
faulty inference	A conclusion that does not follow from the available premises or evidence.	
citation error	A reference that is wrong, missing or does not support the claim attached to it.	

Terms for the connected explanation

Word	Meaning
Atomic claim	A single factual statement checked separately

Word	Meaning
Contradicted	Inconsistent with the supplied evidence
Unsupported	Not established by the available evidence

▼ Earlier wording, alternatives and picture transcripts

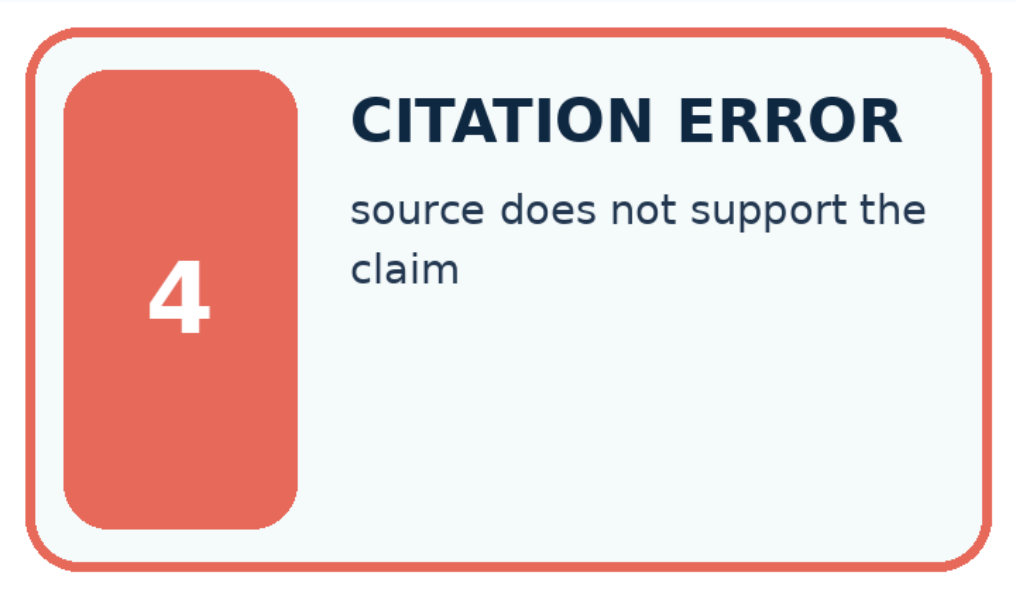
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
FABRICATION
invented detail or source
3
FAULTY INFERENCE
conclusion not justified by evidence



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

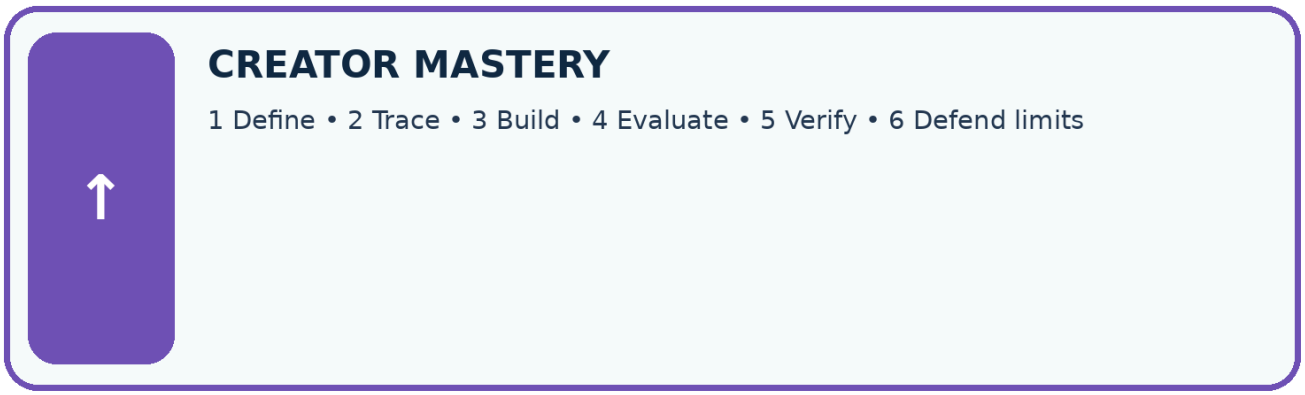
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Audit a new trip summary independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

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Self-check and record

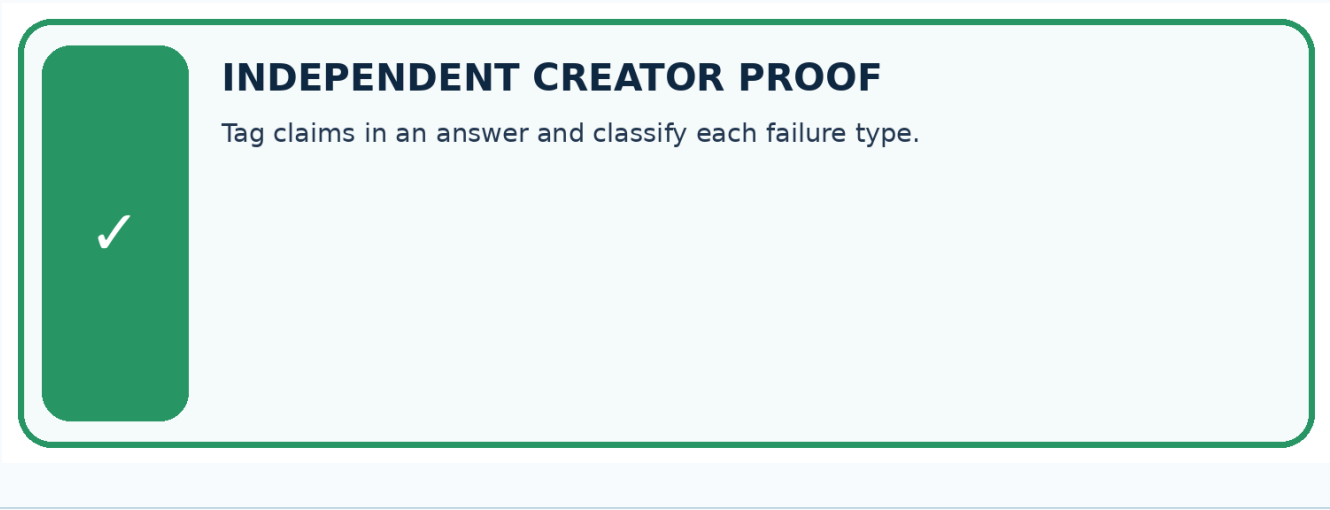
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Audit a new trip summary independently is the selected independent case.



EXPLANATION CHECK Require exact source claims, calculation with units, distinct statuses and limits on observed outcomes.

Write or draw your reasoning and evidence.

▼ **Extra practice: Different failures require different checks**

REFERENCE · SECTION 13

Extra practice: Different failures require different checks

> Try the question before opening its worked explanation.

Read the supplied case and question

An answer invents a quotation, cites a source lacking it and supplies an impossible date.

Separate the failures and checks.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Check exact quote text, claim-to-source support and calendar consistency independently.
Each failure needs its own correction.

Evidence or stage	Value or result
Quotation	Locate exact source text
Citation	Check support for the claim
Date	Check source and calendar consistency
Repair	Correct or remove unsupported statements

▼ Earlier wording, alternatives and picture transcripts

Week 14 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Different failures require different checks

An answer invents a quotation, cites a source lacking it and supplies an impossible date.

Separate the failures and checks.

Check exact quote text, claim-to-source support and calendar consistency independently. Each failure needs its own correction.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of generation failures and verification	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: A fictional Study Buddy drafts a field-trip summary from one approved record.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional Study Buddy drafts a field-trip summary from one approved record.
- Which labelled step matches this detail? Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact.
- 1. **Unsupported claim**
- 2. Supported claim
- 3. **Contradicted** claim
- 4. Compute duration
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: **Unsupported claim** Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact. Different errors need different repairs.

Recalculate the duration; keep the supported departure; remove or seek **evidence** for lunch. These categories are more informative than calling the whole answer wrong.

Arithmetic inherits assumptions such as constant speed and no stops, so avoid turning the three-hour walking calculation into a guaranteed arrival time.

Changed-case question

- Changed-case practice: Add a planned 30-minute stop while preserving speed and distance. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Planned total elapsed time becomes 3.5 hours.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Planned total elapsed time becomes 3.5 hours. Three moving hours plus half an hour stopped; walking-time calculation itself stays three.

Repair and save your work

Review Multimodal **Evidence** (ai-3/13). Goal: Keep visible **evidence**, hidden contents and caption claims separate. Short practice: Why were both “exactly three” and “exactly five” **unsupported** before opening the box?. Return to this week and explain what you changed.

Review Visual Composition (ai-3/11). Goal: Check object counts, spatial relations and arrow meaning separately. Short practice: Which checks pass after that single change, and what still needs repair?. Return to this week and explain what you changed.

Review Structured **Output** and Schemas (ai-3/8). Goal: Separate JSON parsing, schema validation and source support checks. Short practice: Which object demonstrates that valid JSON and valid types are insufficient for truth?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

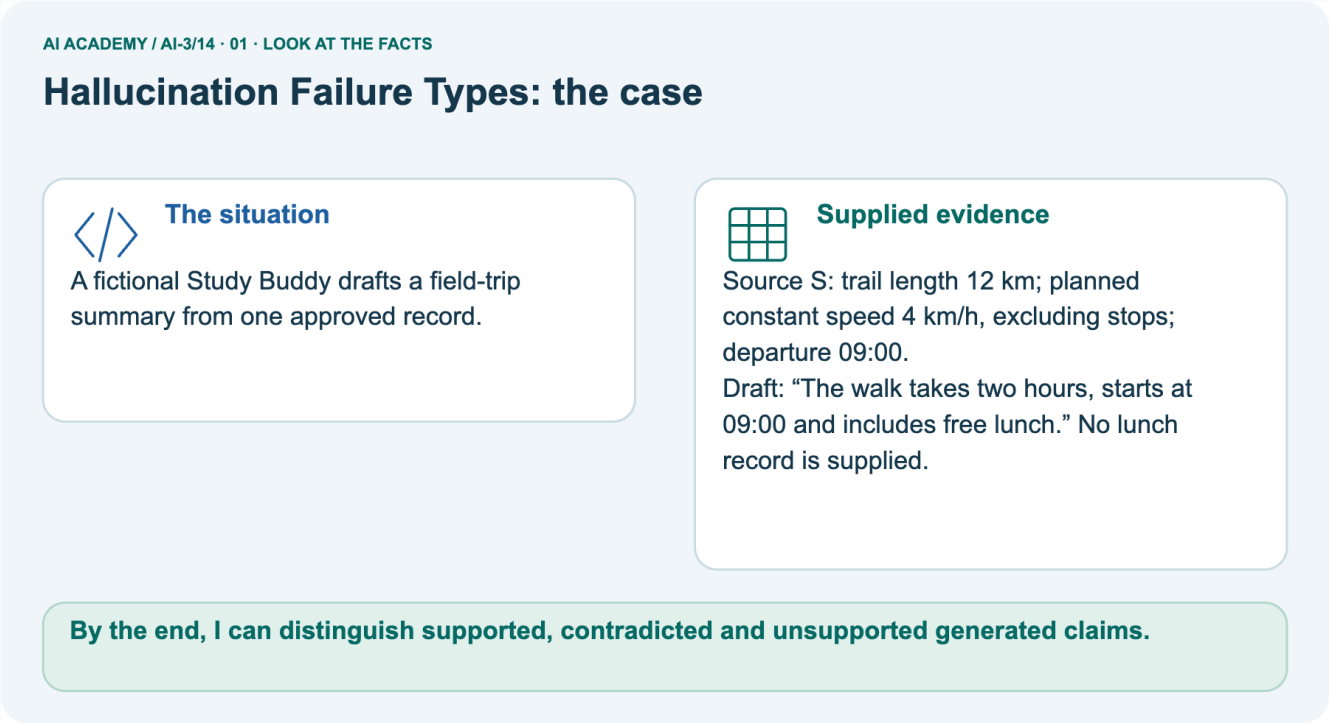
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Hallucination Failure Types: the case



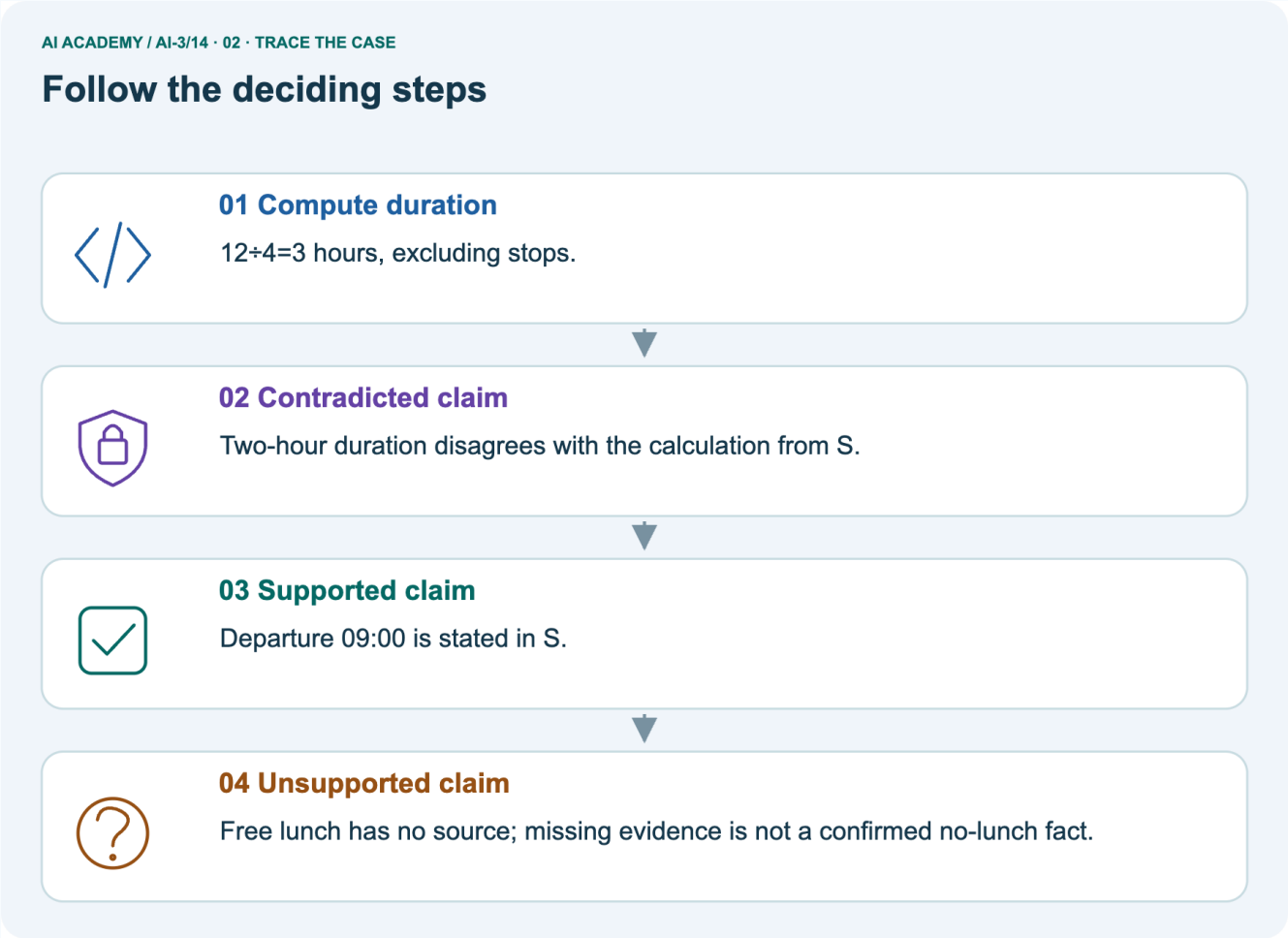
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Study Buddy drafts a field-trip summary from one approved record. Supplied facts: Source S: trail length 12 km; planned constant speed 4 km/h, excluding stops; departure 09:00.

Draft: “The walk takes two hours, starts at 09:00 and includes free lunch.” No lunch record is supplied.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

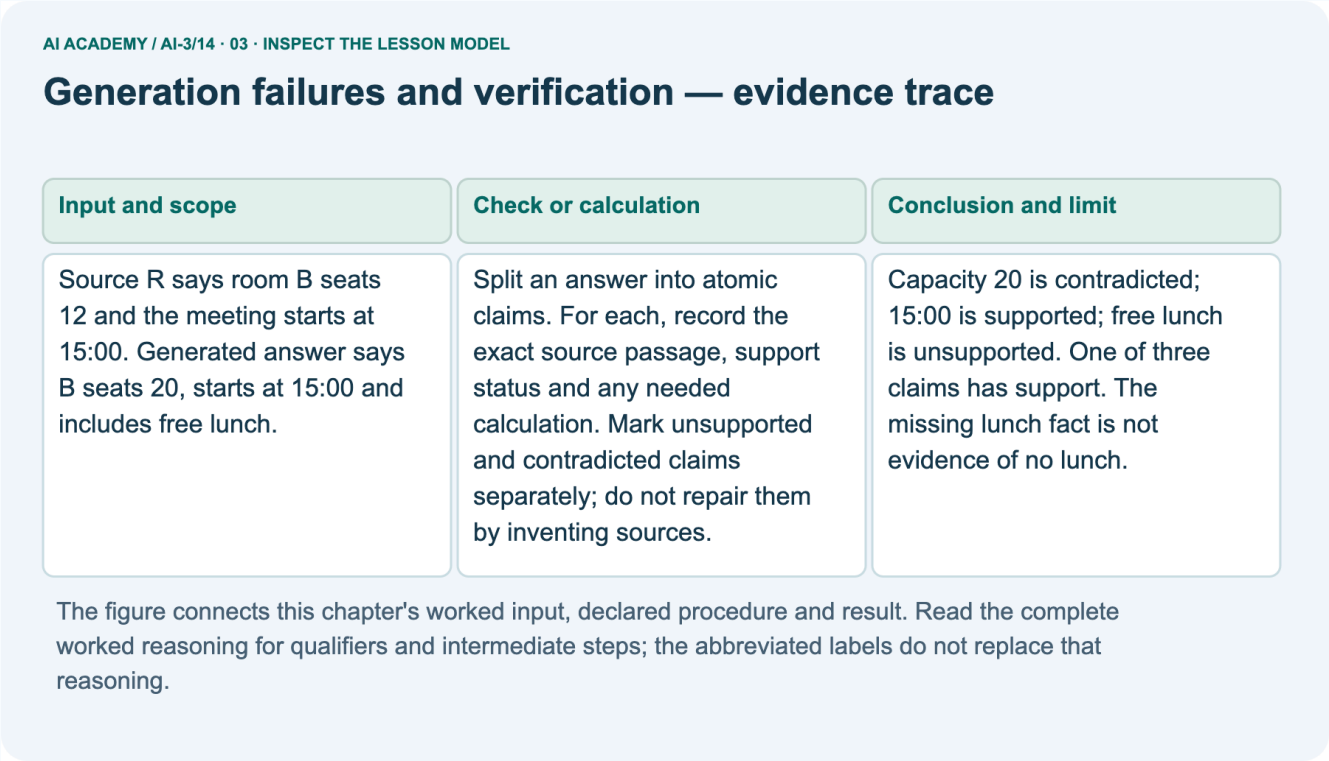


Follow the labelled picture with the matching explanation. Open the vector diagram

Compute duration: 12÷4=3 hours, excluding stops. → **Contradicted** claim: Two-hour duration disagrees with the calculation from S. → Supported claim: Departure 09:00 is stated in S. → **Unsupported claim**: Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact.

Point to each numbered stage and explain which recorded **input** it uses.

03. Generation failures and verification — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Generation failures and verification — **evidence** trace. Source R says room B seats 12 and the meeting starts at 15:00.

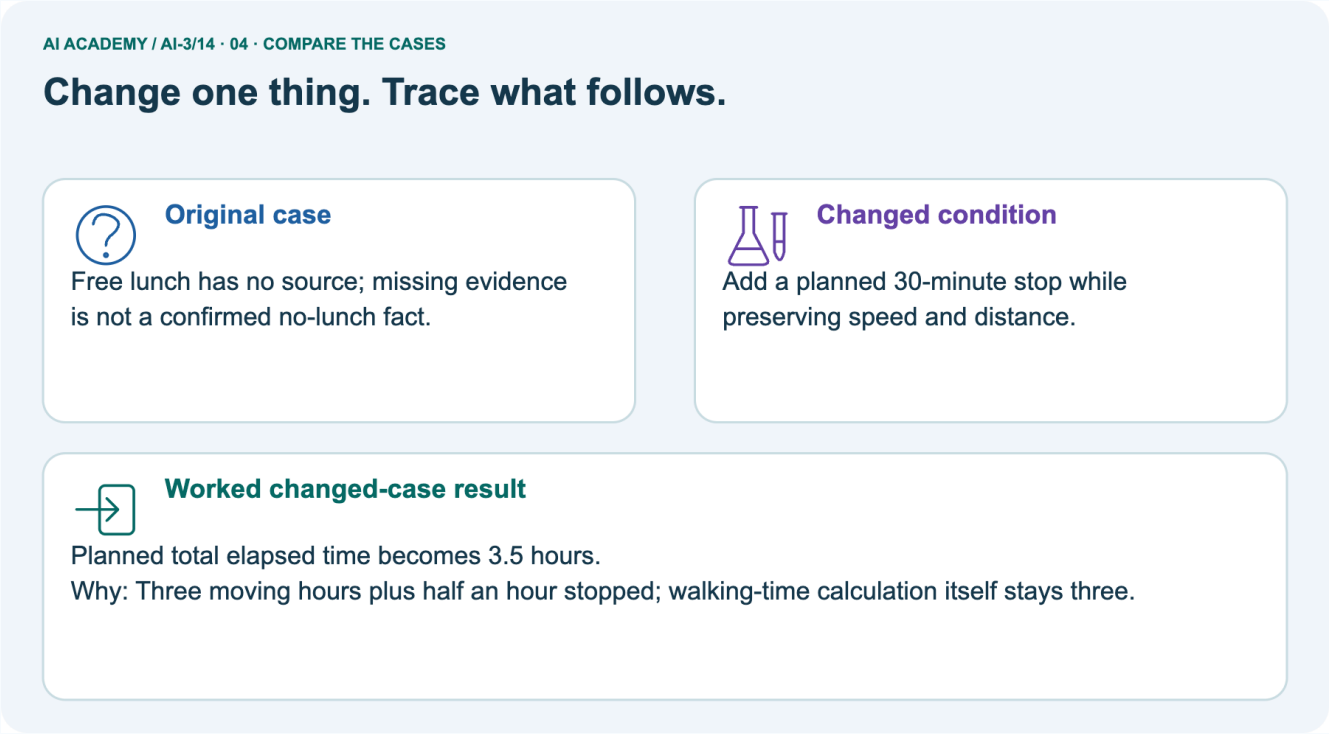
Generated answer says B seats 20, starts at 15:00 and includes free lunch. | Split an answer into **atomic claim**s. For each, record the exact source passage, support status and any needed calculation.

Mark **unsupported** and **contradicted** claims separately; do not repair them by inventing sources. | Capacity 20 is **contradicted**; 15:00 is supported; free lunch is **unsupported**. One of three claims has support.

The missing lunch fact is not **evidence** of no lunch.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



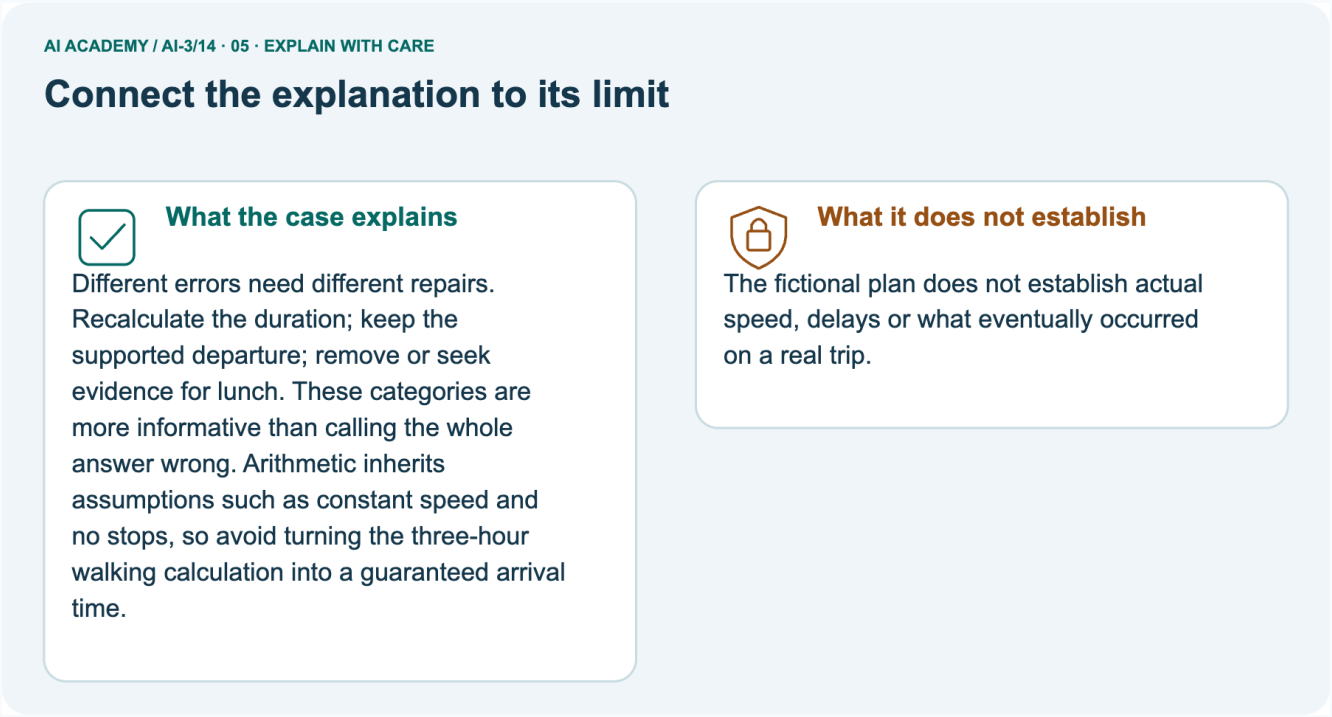
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact. Changed condition: Add a planned 30-minute stop while preserving speed and distance.

Expected result: Planned total elapsed time becomes 3.5 hours. Reason: Three moving hours plus half an hour stopped; walking-time calculation itself stays three.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

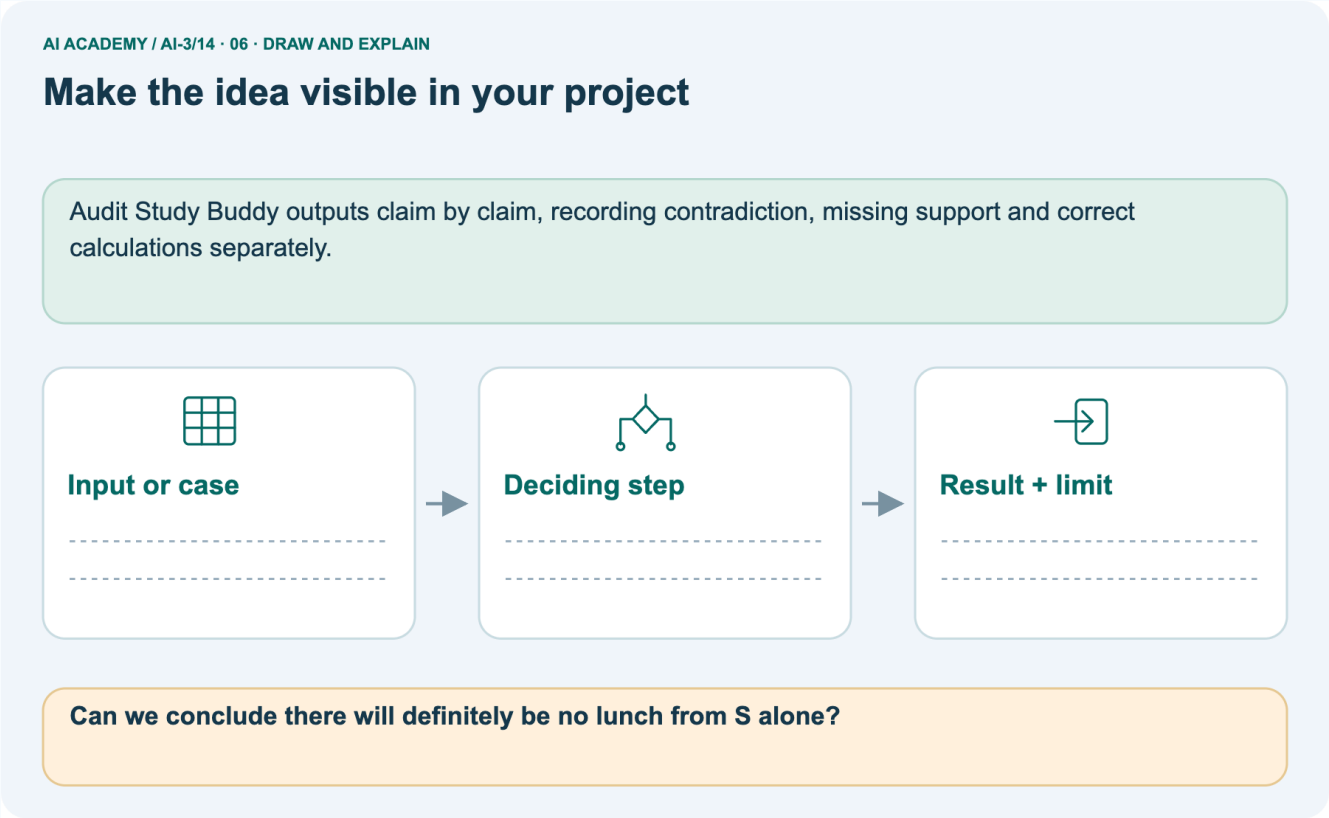
Different errors need different repairs. Recalculate the duration; keep the supported departure; remove or seek **evidence** for lunch. These categories are more informative than calling the whole answer wrong.

Arithmetic inherits assumptions such as constant speed and no stops, so avoid turning the three-hour walking calculation into a guaranteed arrival time.

Limit: The fictional plan does not establish actual speed, delays or what eventually occurred on a real trip.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Audit Study Buddy **output**s claim by claim, recording contradiction, missing support and correct calculations separately. Draw your **input**, deciding step, result and limit.

Supported question: Can we conclude there will definitely be no lunch from S alone?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Hallucination Failure Types: the case

02. Follow the deciding steps

03. Generation failures and verification — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.